Common Loss Functions

Regression & Classification

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Ho Chi Minh City on February 5, 2026

Outline

1. Regression Loss
2. Label Representation for Classification
3. Classification Loss

Regression Loss

Regression Loss

Problem Setup

Given:

$$y \in \mathbb{R}, \quad \hat{y} = f(\boldsymbol{x})$$

Residual:

$$e = y - \hat{y}$$

Goal

Measure how far prediction $\hat{y}$ is from true value y.

Regression Loss

Mean Squared Error (MSE)

$$L_{\text{MSE}} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

- Square makes all errors positive
- Large errors are penalized more

Regression Loss

Step-by-step Example (MSE)

$$\boldsymbol{y} = [3,\ -0.5,\ 2,\ 7]$$

$$\hat{\boldsymbol{y}} = [2.5,\ 0,\ 2,\ 8]$$

i	y_i	$\hat{y}_i$	$(y_i - \hat{y}_i)^2$
1	3	2.5	0.25
2	-0.5	0	0.25
3	2	2	0
4	7	8	1

$$L_{\text{MSE}} = \frac{0.25 + 0.25 + 0 + 1}{4} = 0.375$$

Regression Loss

Mean Absolute Error (MAE)

$$L_{\text{MAE}} = \frac{1}{n} \sum |y_i - \hat{y}_i|$$

Example:

$$|0.5| + |-0.5| + |0| + |-1| = 2$$

$$L_{\text{MAE}} = 2/4 = 0.5$$

Difference

- MAE: linear penalty
- MSE: quadratic penalty

Label Representation for Classification

Label Representation

What is a Label?

In classification, each sample has a class label:

$$y \in \{\text{cat}, \text{dog}, \text{car}, \dots\}$$

Two Views

- Human view: semantic text (cat, dog, ...)
- Machine view: numeric encoding

Label Representation

Text to Index Encoding

Example:

Class name	Index
cat	0
dog	1
horse	2

Now each sample has:

$$y \in \{0, 1, 2\}$$

Important

Index is a **symbol**, not a numeric magnitude.

Label Representation

One-hot Encoding (Hard Label)

For K classes:

$$y \to \boldsymbol{y} \in \{0, 1\}^K$$

Example (dog = class 1):

$$\boldsymbol{y} = [0,\ 1,\ 0]$$

- Exactly one entry is 1
- Others are 0

Meaning

True class has probability 1, others 0

Label Representation

Label as Probability Distribution

One-hot label is a special probability distribution:

$$\sum_{k=1}^{K} y_k = 1$$

Example:

$$[0, 1, 0]$$

means:

$$P(\text{dog}) = 1, \quad P(\text{others}) = 0$$

Key Idea

Classification targets are distributions.

Label Representation

Soft Label

Sometimes labels are not strictly one-hot:

$$\boldsymbol{y} = [0.05,\ 0.90,\ 0.05]$$

- Still sum to 1
- Express uncertainty

Sources

- Label smoothing
- Knowledge distillation
- Human ambiguity

Label Representation

Hard vs Soft Labels

	Hard Label	Soft Label
Form	one-hot	distribution
Uncertainty	none	allowed
Example	[0,1,0]	[0.1,0.8,0.1]

Observation

Both are valid targets for cross-entropy.

From Labels to Loss

We have:

$$\boldsymbol{y} = \text{true distribution}, \quad \hat{\boldsymbol{p}} = \text{predicted distribution}$$

Goal

Measure how different these two distributions are.

Next

Cross-Entropy Loss

Classification Loss

Classification Loss

Key Idea

Model outputs probability:

$$\hat{p} = P(y = 1|\boldsymbol{x})$$

Loss should be:

- Small if model assigns high probability to correct class
- Large if confident but wrong

Binary Cross Entropy (BCE)

For binary $y \in \{0, 1\}$:

$$L_{\text{BCE}} = -\Big[y \log(\hat{p}) + (1 - y) \log(1 - \hat{p})\Big]$$

Two cases:

- If $y = 1$: $L = -\log(\hat{p})$
- If $y = 0$: $L = -\log(1 - \hat{p})$

BCE: Visual Intuition

If true label = 1

$$\hat{p} = 0.9 \Rightarrow L = -\log(0.9) = 0.105$$

$$\hat{p} = 0.1 \Rightarrow L = -\log(0.1) = 2.30$$

Interpretation

Wrong but confident $\Rightarrow$ huge loss

BCE: Step-by-step Example

$$y = 1, \quad \hat{p} = 0.8$$

$$L = -\log(0.8) = 0.223$$

Another sample:

$$y = 0, \quad \hat{p} = 0.7$$

$$L = -\log(1 - 0.7) = -\log(0.3) = 1.204$$

Multiclass Cross Entropy (CE)

K classes, one-hot label:

$$\boldsymbol{y} = (0, 0, 1, 0)$$

Predicted probabilities:

$$\hat{\boldsymbol{p}} = (0.1, 0.2, 0.6, 0.1)$$

$$L_{\text{CE}} = -\sum_{k=1}^{K} y_k \log(\hat{p}_k)$$

CE: Step-by-step

Only true class contributes:

$$L = -\log(0.6) = 0.511$$

If model predicts:

$$\hat{\boldsymbol{p}} = (0.05, 0.05, 0.05, 0.85)$$

$$L = -\log(0.05) = 2.996$$

Meaning

Cross-entropy = negative log probability of correct class

Softmax + CE = Log Loss

Network outputs logits:

$$z_k = \boldsymbol{w}_k^\top \boldsymbol{h} + b_k$$

Softmax:

$$\hat{p}_k = \frac{e^{z_k}}{\sum_j e^{z_j}}$$

CE Loss:

$$L = -\log(\hat{p}_y)$$

Interpretation

Maximize probability of correct class

Why Logarithm?

- Turns multiplication of probabilities into addition
- Strongly penalizes confident mistakes
- Smooth and differentiable

Loss Summary

Regression

- MSE, MAE, RMSE

Classification

- Binary Cross Entropy
- Categorical Cross Entropy

Key Insight

Loss $=$ measure of surprise when seeing the true label